

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: BTech Branch : Electrical Engineering / Electrical Engineering

(Electronics and Power)/ Electrical & Electronics Engg / Electrical & Power Engg

Subject Code & Name: BTEEPE703D Electrical Utilization

Semester: VII

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

Level/CO Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- 1 Which of the following is NOT an advantage of electrical heating?

1

- A) High efficiency B) Quick response C) Low operating cost D) Clean and pollution-free

- 2 What is the primary principle of dielectric heating?

1

- A) Resistance heating B) Magnetic induction C) Dielectric loss in an alternating electric field D) Heat generation by friction

- 3 Which law governs the amount of material deposited during electrolysis?

1

- A) Ohm's Law B) Faraday's Law C) Kirchhoff's Law D) Ampere's Law

- 4 Which type of battery is commonly used in electric vehicles due to its high energy density?

1

- A) Lead-acid battery B) Nickel-cadmium battery C) Lithium-ion battery D) Zinc-carbon battery

- 5 What is the minimum illumination level recommended for general office work?

1

- A) 50 lux B) 100 lux C) 300 lux D) 500 lux

- 6 What is the primary consideration for factory lighting design?

1

- A) Aesthetic appeal B) Energy efficiency and productivity C) Use of decorative luminaires D) Minimized use of light fittings

- 7 Which motor is most commonly used for electric traction?

1

- A) Induction motor B) DC series motor C) Synchronous motor D) Stepper motor

8	What is the main advantage of using a flywheel drive in traction systems?			1
	A) Increased speed	B) Reduction in energy consumption	C) Enhanced braking	D) Reduced noise
9	Which traction system involves using 3-phase alternators to supply DC traction motors?			1
	A) DC generator and series motor	B) Diesel-electric transmission	C) Electric transmission employing 3-phase alternator	D) Battery-operated traction system
10	Which of the following factors affect schedule speed?			1
	A) Type of traction motor	B) Acceleration and deceleration rates	C) Weight of passengers	D) Voltage of supply
11	What is the primary purpose of regenerative braking?			1
	A) Increase the speed of the train	B) Reduce noise during braking	C) Recover and save energy during braking	D) Avoid wear and tear of brakes
12	Which component of train resistance is caused by air friction?			1
	A) Rolling resistance	B) Gradient resistance	C) Wind resistance	D) Mechanical resistance
Q. 2	Solve the following.			12
A)	Explain the working principle of an induction furnace and its applications in industry.	CO1		6
B)	Describe the construction and operating principle of ultrasonic welding. How does it differ from resistance welding?	CO2		6
Q.3	Solve the following.			12
A)	State Faraday's laws of electrolysis and explain their significance in electrolytic processes.	CO1		6
B)	Describe the process of electroplating. What factors influence the quality of electroplating?	CO2		6
Q. 4	Solve Any Two of the following.			12
A)	Explain the procedure for designing a flood lighting scheme for a sports complex.	CO1		6
B)	What are the key requirements of a good lighting scheme? Provide examples for indoor and outdoor applications.	CO2		6
C)	Calculate the illumination at a point 5 meters away from a light source with a luminous flux of 1200 lumens. Assume uniform distribution and neglect other losses.	CO3		6
Q.5	Solve Any Two of the following.			12
A)	Compare the suitability of D.C. series motors and induction motors for electric traction applications.	CO1		6

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|----|---|-----|---|
| B) | Explain the role of a hybrid drive system in modern electric traction. How does it improve efficiency? | CO2 | 6 |
| C) | Describe the construction and working of a trolley bus system. What are its advantages and limitations? | CO3 | 6 |

Q. 6 Solve Any Two of the following. **12**

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| A) | Explain the significance of the speed-time curve in analyzing train movement. | CO1 | 6 |
| B) | Discuss the differences between mechanical braking and regenerative braking. Which is more energy-efficient and why? | CO2 | 6 |
| C) | Calculate the tractive effort required for a train weighing 500 tons to overcome a gradient of 1 in 100. Assume no additional resistances. | CO3 | 6 |

*** End ***